

TECHNICAL REFERENCE

ValidifyPro Admin Portal

Application and Technical Documentation

Features | Architecture | APIs | Data Storage | Configuration | Operations

Repository snapshot

main @ 905a184, including the inspected working-tree state

Prepared 23 July 2026

Scope: This document is derived from the current repository implementation. Credential values, private keys, access tokens, and service-account contents are intentionally excluded.

Document map

Use this page as a fast index. The document distinguishes implemented behavior from installed-but-unused packages and from adjacent Firestore rules.

Section directory

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Quick facts

Application at a glance

Topic	Current implementation
Application	ValidifyPro Admin Portal / Matter Viewer Portal
Framework	Next.js 16.2.4 App Router with React 19.2.4
Primary database	Google Firebase Firestore through Firebase Admin SDK
CRM	Zoho CRM v7; CRM v8 is available for attachment upload
File storage	Zoho WorkDrive; Firestore stores resource metadata and public/share URLs
Supported resource types	Files, links, and notes; logical folders are available in visa templates
Matter identifiers	Firebase application document ID or Zoho Deal ID
Visa experiences	Partner, Protection, Skills in Demand (subclass 482), and Employer Nomination (subclass 186)
UI language	JavaScript and JSX, with server-side Node.js route handlers
Current access posture	Matter and admin route handlers are not effectively access-gated in the current source

1. Application overview and architecture

The portal gives staff a consolidated view of an immigration matter while keeping operational records in Firestore and file content in Zoho WorkDrive.

1.1 Purpose

The application is a Next.js-based matter review and resource administration portal. A staff member enters either a Firebase Application ID or a Zoho Deal ID, then reviews questionnaire data, manages matter-specific and shared resources, uploads a prepared PDF for document review, and tracks correction status.

- **Matter review:** Resolve one matter, show application metadata, and calculate questionnaire completion.
- **Questionnaire review:** Present stored answers as read-only fields with profile-aware navigation.
- **Reviewer notes:** Create and resolve field-level notes stored beneath the application.
- **Resource management:** Manage matter files, notes, links, global legacy resources, and visa-specific resource templates.
- **Document review:** Upload and preview a PDF, then reconcile issues from Firestore and Zoho CRM Corrections.

1.2 System boundary

Architecture: Browser UI -> Next.js App Router pages and route handlers -> Firestore for structured records and metadata; Zoho CRM for Deal fields and Corrections; Zoho WorkDrive for uploaded file bytes and share links.

The browser does not directly query Firestore. Pages call local /api route handlers. Those handlers use the Firebase Admin SDK, which operates with service-account privileges and bypasses Firestore security rules. Zoho calls are also server-side through a shared Axios client.

1.3 Runtime components

Logical architecture

Layer	Responsibility
Presentation	Client components under <code>src/app</code> and <code>src/components</code> render the landing page, matter workspace, questionnaire, resources, and admin dashboards.
Application API	Next.js route handlers validate requests, resolve matter IDs, serialize Firestore timestamps, and coordinate external writes.
Domain helpers	<code>src/lib</code> contains matter resolution, questionnaire structure/progress, resource normalization, session helpers, and the Zoho client.
Persistent records	Firestore stores applications, questionnaires, completion maps, comments, notifications, resource

Layer	Responsibility
	metadata, and template metadata.
External content	Zoho WorkDrive stores uploaded files; Zoho CRM stores Deal data, final-document URLs, and Corrections records.

1.4 Matter identity and duplicate handling

1. Attempt a direct Firestore lookup at applications/{matterId}.
2. If no direct document exists, query zohold, then zohoDealId, then dealId for an exact match.
3. If more than one application matches, hydrate questionnaire and completion data for each candidate.
4. Choose the best candidate by most recent timestamp, then completed-key count, profile count, data-document presence, and finally application ID.
5. Return the canonical Firebase application ID so the matter layout can replace a Zoho-ID URL with a Firebase-ID URL.

2. Feature catalog and user journeys

This section describes behavior visible in the current source, including differences between the generic questionnaire review and the subclass 482/186 review.

2.1 Landing page and matter workspace

- Search by Firebase Application ID or Zoho Deal ID.
- Redirect directly to the matter questionnaire.
- Show reference, normalized visa label, Firebase ID, Zoho Deal ID, and completion percentage.
- Use a sticky responsive header with Questionnaire, Resources, and Document Review tabs.
- Collapse the matter summary after scrolling while keeping navigation visible.
- Display database and not-found errors with a return-to-search action.

2.2 Questionnaire review

Questionnaire data is read from applications/{appId}/data/questionnaire. The UI selects one of two review experiences:

Questionnaire review modes

Mode	Implemented behavior
Skills in Demand / Employer Nomination	Triggered by temporary-work, visaContext 482/186, or matching labels. Provides profile-aware sidebar navigation, completion markers, read-only field controls, Previous/Next movement, included-applicant summaries, and all-applicant sections.
Generic structured review	Used for other matters. Builds applicant, all-applicant, non-migrating, and uncategorized sections. Provides profile/section tabs, field-level reviewer notes, and browser print export with an option to include notes.

- **Main applicant topics:** Details, other names, identity, contact details, employment, education, skills, and language.
- **Spouse topics:** Details, other names, and identity; subclass 186 also adds education and language.
- **Child topics:** Details, other names, identity, and custody.
- **All-applicant topics:** Visas, travel history, countries of residence, health, and character.
- **Progress:** Temporary-work progress is calculated against profile-aware expected sections; other visa types use the configured route catalog.

2.3 Reviewer comments

- Create a comment against a field path with label, body, severity, source, and optional document URL.

- Store open/resolved status, a derived section key, author placeholders, and timestamps.
- List comments in creation order and optionally filter by source.
- Edit body or severity and change status between open and resolved.
- Delete Firestore comments.
- When source=documentReview, merge Zoho CRM Corrections into the read response and update their status in Zoho when changed.

2.4 Matter-specific resources

- Create files, links, or notes for one application.
- Search the active matter resource list by title, description, note content, file name, or URL.
- Upload files up to 50 MB to the WorkDrive folder referenced by the Zoho Deal's Workdrive_Folder_ID or legacy WorkDrive_Folder_ID field.
- Store resource metadata under the application in Firestore.
- Archive a resource; if it has a WorkDrive resource ID, the external resource is moved to WorkDrive deleted status before Firestore is marked archived.
- Embed, Library, and Folder buttons are shown as disabled future options on the individual-resource screen.

2.5 Visa-specific shared resource templates

The All Matters tab embeds the template manager. Templates are seeded on demand for four visa slugs and use fixed WorkDrive root folders with environment-variable overrides.

Resource templates

Visa slug	Display title	WorkDrive ownership
partner	Subclass 820/801	Partner template root
protection	Subclass 866	Protection template root
482	Subclass 482	Skills in Demand template root
186	Subclass 186	Employer Nomination template root

- View all templates or filter to one visa type.
- Create named categories with an icon; categories are stored on the template document.
- Create file, link, note, and logical folder items.
- Upload multiple files and retain the public link returned by WorkDrive.
- Search, sort, edit metadata, hide/reactivate, move between logical parents, and delete items.
- Reject missing/non-folder parents and prevent folder cycles.

- Delete WorkDrive content before deleting a file item from Firestore.

Important: Template folder items and categories are logical Firestore organization. File uploads go to the configured visa root folder; the current implementation does not create corresponding WorkDrive subfolders.

2.6 Document review

- Accept only PDF uploads for the document-review source, verifying both the .pdf extension and the %PDF- file signature.
- Enforce a 50 MB limit.
- Find or create a matter-named folder beneath the configured document-review WorkDrive root.
- Upload the file, create a public WorkDrive link, derive an embed URL, and preview it in an iframe.
- Store the resource under the application and copy Final_File_For_Visa_Submission to both Firestore and the Zoho Deal.
- Show open and resolved issues from Firestore plus Zoho Corrections.
- Archive the current review resource and clear the final-file field in Firestore and Zoho.

2.7 Admin dashboards

Admin-facing features

Page	Purpose
/admin/document-review	Aggregate Firestore document-review comments across applications; filter/search and resolve/reopen.
/admin/resources	Legacy global resource library with file/link/note CRUD, statuses, scopes, metadata filters, and quick publishing actions.
/login	Admin-key form exists, but current getSession behavior redirects immediately because it always returns an admin actor.

3. Application pages and technical stack

3.1 Page routes

User-interface route map

Route	Screen	Responsibility
/	Matter lookup	Enter a Firebase application ID or Zoho Deal ID.
/matter/[matterId]	Redirect	Redirects to the questionnaire.
/matter/[matterId]/questionnaire	Questionnaire review	Read answers, progress, and supported reviewer notes/export.
/matter/[matterId]/resources	Resources	Manage visa templates or resources limited to one matter.
/matter/[matterId]/document-review	Document review	Upload/preview PDF and manage issue status.
/admin	Redirect	Redirects to the legacy resource library.
/admin/resources	Legacy library	Global resource management.
/admin/document-review	Corrections dashboard	Cross-matter Firestore correction list.
/login	Admin login	Admin-key form; currently bypassed by the session helper.

3.2 Core technology

Technology stack

Area	Technology	Role
Framework	Next.js 16.2.4	App Router pages and route handlers; dynamic params are awaited.
UI runtime	React 19.2.4 / React DOM 19.2.4	Client-side state, effects, memoization, and responsive views.
Language	JavaScript, JSX, and ESM	No TypeScript source; @/* maps to ./src/*.
Styling	Tailwind CSS 4 + @tailwindcss/postcss	Utility-first layout and the project color system.
UI primitives	Radix UI	Collapsible, Progress, Scroll Area, Slot, and related local wrappers.
Icons	lucide-react 1.11	Interface icons.
Class utilities	clsx, tailwind-merge, class-variance-authority	Conditional classes and component variants.
Database SDK	firebase-admin 13.8	Server-side Firestore and Auth service initialization.
HTTP client	Axios 1.15.2	Zoho token, CRM, and WorkDrive requests.
Multipart	form-data 4.0.5	Server-side WorkDrive and CRM attachment uploads.
Quality	ESLint 9 + eslint-config-next 16.2.4	Next core-web-vitals lint configuration.
Hosting metadata	Firebase configuration present	Firestore region asia-east2, rules, and indexes are tracked.

3.3 Installed but not active in the inspected application flow

- The Firebase client package is installed, but application data access in src uses firebase-admin on the server.

- date-fns and @radix-ui/react-tabs are declared dependencies but are not imported by current src files.
- DATABASE_URL and CUSTOM_DATABASE_URL are present in local environment files but are not referenced by application source.

4. Internal API reference

All endpoints are Next.js route handlers. JSON responses use a success boolean and usually include an error message on failure.

Complete internal endpoint catalog

Method	Route	Purpose
POST	/api/admin/session	Validate PORTAL_ADMIN_KEY and set a signed 12-hour HTTP-only cookie.
DELETE	/api/admin/session	Clear the admin session cookie.
GET	/api/admin/document-review	Collection-group query for Firestore reviewComments where source=documentReview.
GET	/api/matter/[matterId]	Resolve the matter, read application/questionnaire/completion, and calculate progress.
GET	/api/matter/[matterId]/resources	List non-archived resources beneath the resolved application.
POST	/api/matter/[matterId]/resources	Create a file, link, or note; files go to WorkDrive.
PATCH	/api/matter/[matterId]/resources/[resourceId]	Archive a resource; delete external WorkDrive content when an ID is present.
GET	/api/resources	List/filter/search the legacy global resource collection.
POST	/api/resources	Create a global file, link, or note.
PATCH	/api/resources/[resourceId]	Edit global resource metadata, URL/note content, scope, or status.
DELETE	/api/resources/[resourceId]	Delete the Firestore global resource document.
GET	/api/resource-templates	Ensure the four templates exist and return item counts.
GET	/api/resource-templates/[visaSlug]	Return one template and its sorted items.
PATCH	/api/resource-templates/[visaSlug]	Update title, status, or categories.
POST	/api/resource-templates/[visaSlug]/items	Create folder/file/link/note item and update the template timestamp.
PATCH	/api/resource-templates/[visaSlug]/items/[itemId]	Edit name, parent, category, order, status, link, or note.
DELETE	/api/resource-templates/[visaSlug]/items/[itemId]	Delete WorkDrive resource when present, then delete Firestore item.
GET	/api/review-comments/[matterId]	List Firestore comments; source=documentReview also merges Zoho Corrections.
POST	/api/review-comments/[matterId]	Create a Firestore review comment.
PATCH	/api/review-comments/[matterId]/[commentId]	Update a Firestore comment or Zoho Correction status.
DELETE	/api/review-comments/[matterId]/[commentId]	Delete a Firestore review comment.

4.1 Request and validation rules

Boundary validation

Input	Rule
Matter ID	Required; Firebase ID is tried first, followed by exact Zoho-ID fields.
Matter files	Maximum 50 MB; arbitrary type except document-review uploads.
Document-review file	Maximum 50 MB; .pdf extension and PDF header are both required.
Links	Only http and https URLs are accepted.

Input	Rule
Notes	Non-empty note text is required.
Template visa slugs	partner, protection, 482, and 186, with limited aliases normalized.
Template parent	Must be a folder in the same template; folder moves cannot form a cycle.
Comment status	Only open or resolved.
Resource archive	Matter-specific resource PATCH supports only status=archived.

4.2 Response and error behavior

- 400 for missing or invalid request values.
- 401 only where explicit session/key checks run; current session helper normally returns an admin actor.
- 404 for unresolved matters, resources, templates, or items.
- 500 for initialization and unexpected server errors.
- 502 where Zoho/WorkDrive cannot supply required remote data or a usable public link.
- Firestore timestamps are converted to ISO strings before responses are serialized.

5. External APIs and integrations

5.1 Firebase Admin SDK

src/lib/firebase-admin.js initializes a singleton Firebase app on module load. Credential discovery is ordered for hosting environments first and local files last.

1. Decode FIREBASE_SERVICE_ACCOUNT_BASE64, if present.
2. Parse FIREBASE_SERVICE_ACCOUNT_KEY as raw JSON.
3. Load GOOGLE_APPLICATION_CREDENTIALS or FIREBASE_SERVICE_ACCOUNT_PATH.
4. Fallback to service.json or service-account.json in the repository root.
5. Use NEXT_PUBLIC_FIREBASE_PROJECT_ID or project_id from the service account.

Security: The service account grants server-side administrative access. The service JSON is ignored by Git and must not be distributed with application documentation or client bundles.

5.2 Zoho access-token broker

The application does not perform an OAuth refresh-token exchange itself. It performs an HTTP GET to ACCESSTOKEN_URL or ZOHO_ACCESS_TOKEN_URL; WorkDrive can use WORKDRIVE_ACCESSTOKEN_URL first. The broker may return access_token, details.output, output, or a plain token string.

- The Zoho-oauthtoken prefix is stripped when received and added to Authorization headers on outbound calls.
- CRM and WorkDrive tokens are cached independently in memory for 50 minutes.
- CRM requests retry once with a refreshed token after HTTP 401.
- The datacenter is controlled by ZOHO_DATACENTER and defaults to com.au.

5.3 Zoho CRM APIs

Zoho CRM endpoints used or exposed by the shared client

Capability	Endpoint pattern	Portal responsibility
CRM v7 base	https://www.zohoapis.{datacenter}/crm/v7	Record search/CRUD, COQL, and related lists.
Deals	GET /Deals/{id}	Read WorkDrive folder fields and general Deal data.
Deals update	PUT /Deals	Set or clear Final_File_For_Visa_Submission.
Corrections	GET/PUT /Deals/{id}/Corrections	Read document-review issues and update open/resolved status.
CRM v8 attachments	POST /crm/v8/{module}/{id}/Attachments	Generic attachment helper available in the client wrapper.

The shared Zoho client also contains generic contact search, record creation/update, related-record CRUD, COQL, and Partner_Dependents synchronization helpers. Those helpers are available infrastructure but are not called by the current portal route set except for Deals and Corrections operations.

5.4 Zoho WorkDrive API

WorkDrive operations

Operation	Endpoint	Use
List children	GET /files/{folderId}/files	Find an existing matter folder; scans in 50-item pages up to 1,000 children.
Create folder	POST /files	Create the document-review matter folder.
Upload file	POST /upload	Store file bytes in a configured folder.
Create public link	POST /links	Create an external link with download allowed.
Delete resource	PATCH /files/{resourceId}	Set WorkDrive status 51; 404 is treated as already missing.

Data exposure: Uploaded resource URLs are public/external WorkDrive links. Treat the returned URL as shareable access and apply the organization's WorkDrive policy accordingly.

6. Data storage and information ownership

Structured data and file content are intentionally split. Firestore is the system of record for portal metadata; WorkDrive owns file bytes; Zoho CRM owns Deal and Corrections records.

6.1 Firestore collections used by application code

Application-owned Firestore paths

Path	Purpose	Representative content
applications/{appld}	Core matter metadata	Reference, applicant/client fields, visa type, Zoho IDs, timestamps, and final document URL.
applications/{appld}/data/questionnaire	Questionnaire answers	Profiles, profile data, all-applicant answers, visaContext, and legacy keys.
applications/{appld}/data/completion	Progress map	Boolean completion keys and timestamps.
applications/{appld}/reviewComments/{commentId}	Review comments	Path, label, body, severity, source, status, section key, author placeholders, and timestamps.
applications/{appld}/resources/{resourceId}	Matter resources	File/link/note metadata, URLs, WorkDrive IDs, status, source, category, and timestamps.
notifications/{notificationId}	Applicant notification intent	Application ID, type, title/body, path, read flag, and createdAt.
resources/{resourceId}	Legacy global library	Type, title/content, scope, status, category/program/audience, WorkDrive metadata, and lifecycle timestamps.
resourceTemplates/{visaSlug}	Template header	Title, status, categories, WorkDrive root, actor metadata, and timestamps.
resourceTemplates/{visaSlug}/items/{itemId}	Template items	Logical parent, kind, name, category, order, visibility, content/URL, WorkDrive metadata, and timestamps.

6.2 Collections present in Firestore rules but not used by current portal source

- users/{userId} and nested preferences, drafts, and zoho documents.
- messages/{messageId}.
- Authenticated-owner and publicReviewAccess rules for applications and their subcollections.

These rules appear to support an adjacent applicant/client portal. The admin portal uses Firebase Admin on the server, so these client security rules do not restrict its route handlers.

6.3 Zoho CRM ownership

Zoho CRM data used by the portal

Zoho location	Stored information
Deals record	Matter/CRM identity and fields used for WorkDrive folder lookup.
Workdrive_Folder_ID / WorkDrive_Folder_ID	Folder ID for normal matter-specific file uploads.
Final_File_For_Visa_Submission	Preview/embed URL for the current review document; updated in both Zoho

Zoho location	Stored information
	and Firestore.
Corrections related list	Client-submitted document issues, including field/name, issue description, status, email, and timestamps.

6.4 WorkDrive ownership

- Actual bytes for matter-specific files.
- Actual bytes for global legacy-library files.
- Actual bytes for visa-template files.
- Document-review PDFs in matter-named folders beneath the review root.
- External link objects and downloadable/public URLs.

6.5 Browser and process storage

Non-database storage

Location	Key/data	Purpose
Cookie	vp_admin_session	Signed HTTP-only, SameSite=Lax, 12-hour maximum age; secure in production.
React state	Ephemeral UI state	Filters, selected resources, form values, open drawers, and current navigation.
Server memory	Zoho token cache	Separate CRM and WorkDrive access tokens for up to 50 minutes.
Environment/files	Credentials and configuration	Server-only environment variables or ignored service-account JSON files.

7. Core data flows

7.1 Open a matter

1. The landing page navigates to `/matter/{input}/questionnaire`.
2. The matter layout calls `GET /api/matter/{input}`.
3. The route resolves the Firebase application, reads questionnaire and completion documents, and calculates progress.
4. The layout replaces a Zoho-ID URL with the canonical Firebase ID when needed.
5. Child pages call their own APIs for comments, resources, or document-review data.

7.2 Upload a normal matter resource

1. Resolve the application and derive its Zoho Deal ID.
2. Fetch the Deal and read `Workdrive_Folder_ID` or `WorkDrive_Folder_ID`.
3. Upload bytes to WorkDrive and create a public link.
4. Store metadata and WorkDrive identifiers beneath `applications/{appId}/resources`.
5. Return the serialized resource to the browser.

7.3 Upload a document-review PDF

1. Verify the 50 MB limit, .pdf extension, and PDF signature.
2. Resolve the configured review root and find or create a matter-named WorkDrive folder.
3. Upload the PDF and create a public link plus embed URL.
4. Update the Zoho Deal's `Final_File_For_Visa_Submission` field.
5. Commit the Firestore resource and application final-file field in one batch.
6. Preview the document and merge Firestore document-review comments with Zoho Corrections.

7.4 Publish a visa-template file

1. Ensure `resourceTemplates/{visaSlug}` exists and has the current root-folder metadata.
2. Validate kind, status, category, order, and logical parent.
3. Upload file bytes directly to the configured visa WorkDrive root.
4. Create a public link and store the item under `resourceTemplates/{visaSlug}/items`.
5. Update the template's timestamp and actor metadata.

7.5 Resolve an issue

- **Firestore comment:** PATCH updates status and updatedAt on the comment document.
- **Zoho correction:** IDs prefixed zohoCorrection: are translated to a Zoho related-record update with Status=Resolved or Open.

7.6 Deletion and archive semantics

Lifecycle behavior

Object	Current behavior
Matter resource	External WorkDrive resource is deleted when possible; Firestore document remains with status=archived.
Document-review resource	Same as matter resource, plus final-file fields are cleared in Firestore and Zoho.
Template item	WorkDrive resource is deleted first, then the Firestore item is permanently deleted.
Legacy global resource	Firestore document is permanently deleted; the route does not delete the corresponding WorkDrive file.
Review comment	Firestore comment is permanently deleted; Zoho-prefixed corrections are not handled by DELETE.

8. Authentication, authorization, and security

8.1 Implemented session mechanism

The repository includes an admin-key login mechanism. A valid `PORTAL_ADMIN_KEY` is compared with `timingSafeEqual`, then a `role=admin` payload is signed with HMAC-SHA256 using `SESSION_SECRET`. The cookie is HTTP-only, `SameSite=Lax`, secure in production, and expires after 12 hours.

Current behavior: `getAdminSession()` does not read or verify the cookie. It always returns a fresh admin actor, and `requireAdminSession()` delegates to it. Therefore `/login` redirects immediately and routes that call these helpers are effectively open.

8.2 Route access posture

Current authorization posture

Area	Enforcement
Matter pages and <code>/api/matter/*</code>	No authentication check.
<code>/api/review-comments/*</code>	No authentication check.
<code>/api/admin/document-review</code>	Calls <code>requireAdminSession</code> , which currently auto-grants.
<code>/api/resources*</code>	Uses <code>getAdminSession</code> only to label the actor; it currently auto-grants.
<code>/api/resource-templates*</code>	Checks <code>getAdminSession</code> , which currently auto-grants.
<code>/api/admin/session</code>	POST validates the admin key, but the rest of the app does not require the resulting cookie.

8.3 Security controls that are present

- Server-only Firebase and Zoho credentials.
- HMAC-signed session token implementation and timing-safe key comparison.
- HTTP/HTTPS URL allow-list for submitted links.
- File-name sanitization for WorkDrive uploads.
- 50 MB upload limits.
- Document-review PDF signature verification.
- Template parent validation and folder-cycle prevention.
- No service-account or `.env` files tracked by the repository ignore rules.

8.4 Security actions before external production exposure

- Restore real cookie verification in `getAdminSession` and `requireAdminSession`.
- Apply the gate consistently to matter reads and every mutation endpoint, or implement a separate authenticated user role model.

- Add CSRF protection or an equivalent same-origin mutation strategy once sessions are enforced.
- Review whether public WorkDrive links are appropriate for each resource category.
- Avoid logging token-response bodies or record samples in production.
- Confirm client-portal Firestore rules for resources, resourceTemplates, notifications, and reviewComments if clients will access Firestore directly.

9. Configuration and environment variables

Values are intentionally omitted. Configure these in the deployment environment; do not place secrets in NEXT_PUBLIC variables.

Environment variables referenced by source

Variable	Area	Purpose
NEXT_PUBLIC_FIREBASE_PROJECT_ID	Firebase	Required unless project_id is supplied by service account.
FIREBASE_SERVICE_ACCOUNT_BASE64	Firebase	Preferred hosted credential option; base64 JSON.
FIREBASE_SERVICE_ACCOUNT_KEY	Firebase	Raw service-account JSON alternative.
GOOGLE_APPLICATION_CREDENTIALS	Firebase	Path to service-account file.
FIREBASE_SERVICE_ACCOUNT_PATH	Firebase	Application-specific path alternative.
FIREBASE_PROJECT_ID	Migration script	Project ID fallback used by scripts/migrate-shared-resources.js.
ACCESSTOKEN_URL	Zoho	Primary token-broker URL.
ZOHO_ACCESS_TOKEN_URL	Zoho	Backward-compatible token-broker URL.
WORKDRIVE_ACCESSTOKEN_URL	Zoho	Optional WorkDrive-specific token broker.
ZOHO_DATACENTER	Zoho	Optional; defaults to com.au.
PORTAL_ADMIN_KEY	Session	Admin-key endpoint secret.
SESSION_SECRET	Session	HMAC secret for signed cookie.
DOCUMENT_REVIEW_WORKDRIVE_FOLDER_ID	WorkDrive	Document-review root override.
WORKDRIVE_DOCUMENT_REVIEW_FOLDER_ID	WorkDrive	Legacy/alternate review root override.
SHARED_RESOURCES_WORKDRIVE_FOLDER_ID	WorkDrive	Preferred global legacy-library root override.
WORKDRIVE_SHARED_FOLDER_ID	WorkDrive	Global root fallback.
SHARED_WORKDRIVE_FOLDER_ID	WorkDrive	Global root fallback.
WORKDRIVE_FOLDER_ID	WorkDrive	Broad global root fallback.
RESOURCE_TEMPLATE_PARTNER_WORKDRIVE_FOLDER_ID	Templates	Partner template root override.
PARTNER_RESOURCE_TEMPLATE_WORKDRIVE_FOLDER_ID	Templates	Partner alternate override.
RESOURCE_TEMPLATE_PROTECTION_WORKDRIVE_FOLDER_ID	Templates	Protection template root override.
PROTECTION_RESOURCE_TEMPLATE_WORKDRIVE_FOLDER_ID	Templates	Protection alternate override.
RESOURCE_TEMPLATE_482_WORKDRIVE_FOLDER_ID	Templates	Subclass 482 root override.
VISA_482_RESOURCE_TEMPLATE_WORKDRIVE_FOLDER_ID	Templates	Subclass 482 alternate override.
RESOURCE_TEMPLATE_186_WORKDRIVE_FOLDER_ID	Templates	Subclass 186 root override.
VISA_186_RESOURCE_TEMPLATE_WORKDRIVE_FOLDER_ID	Templates	Subclass 186 alternate override.

9.1 Present locally but not used by current application source

Configured-but-unreferenced variable names

Variable	Status
NEXT_PUBLIC_FIREBASE_API_KEY	Configured locally; no src reference.
NEXT_PUBLIC_FIREBASE_APP_ID	Configured locally; no src reference.
NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN	Configured locally; no src reference.
NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID	Configured locally; no src reference.
NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET	Configured locally; no src reference.
DATABASE_URL	Configured locally; no src reference.
CUSTOM_DATABASE_URL	Configured locally; no src reference.

10. Folder structure and source ownership

Generated folders such as `node_modules` and `.next` are omitted. The tree below focuses on maintained source and operational configuration.

```

validify-pro-admin-portal/
|-- public/                               Static logos, favicon, and starter SVG assets
|-- scripts/
|   |-- migrate-shared-resources.js        Matter-resource to global-library migration
|-- src/
|   |-- app/
|   |   |-- admin/                        Admin layouts and dashboard pages
|   |   |-- api/                          Next.js route handlers
|   |   |-- login/                        Admin-key form route
|   |   |-- matter/[matterId]/            Matter shell, questionnaire, resources, review
|   |   |-- globals.css                   Tailwind import and print/responsive styling
|   |   |-- layout.js                     Root metadata and HTML shell
|   |   |-- page.js                       Matter lookup landing page
|   |-- components/
|   |   |-- admin/                        Admin managers and session controls
|   |   |-- ui/                           Local reusable UI primitives
|   |   |-- QuestionnaireSidebar.jsx
|   |   |-- SkillsInDemandQuestionnaireReview.jsx
|   |-- lib/                              Firebase, Zoho, domain rules, and helpers
|-- .env / .env.local                     Local configuration; ignored by Git
|-- firebase.json                         Firestore deployment configuration
|-- firestore.rules                       Client Firestore security rules
|-- firestore.indexes.json                Messages indexes
|-- next.config.mjs                       Next.js configuration
|-- package.json                          Dependencies and npm scripts
|-- postcss.config.mjs                    Tailwind PostCSS plugin
|-- jsconfig.json                         @/* -> src/* import alias

```

10.1 API route ownership

```

src/app/api/
|-- admin/
|   |-- document-review/route.js          Cross-application Firestore corrections
|   |-- session/route.js                  Admin-key cookie create/delete
|-- matter/[matterId]/
|   |-- route.js                          Matter + questionnaire + progress
|   |-- resources/
|   |   |-- route.js                      List/create matter resources
|   |   |-- [resourceId]/route.js         Archive matter resource
|-- resource-templates/
|   |-- route.js                          Seed/list templates
|   |-- [visaSlug]/
|   |   |-- route.js                      Get/update template
|   |   |-- items/
|   |   |   |-- route.js                  Create template item
|   |   |   |-- [itemId]/route.js         Update/delete item
|-- resources/
|   |-- route.js                          Legacy global list/create
|   |-- [resourceId]/route.js             Legacy update/delete
|-- review-comments/[matterId]/
|   |-- route.js                          List/create comments + Zoho merge
|   |-- [commentId]/route.js              Update/delete comments

```

10.2 Key libraries

Library responsibilities

File	Responsibility
adminSession.js	Admin-key verification, HMAC token creation, cookie options, and current auto-grant helpers.
firebase-admin.js	Credential discovery and singleton Firebase Admin/Firestore initialization.
matterResolver.js	Firebase/Zoho ID resolution and duplicate selection.
pdfUploadRules.mjs	PDF header and extension validation.
questionnaireProgress.js	Profile-aware temporary-work progress calculation.
questionnaireSections.js	Convert raw questionnaire keys into applicant/shared/non-migrating sections.
resourceTemplates.js	Template definitions, normalization, Firestore seed/read helpers, WorkDrive upload/delete.
routes.js	Partner, protection, subclass 482, and subclass 186 intake route catalogs.
sharedResources.js	Global resource normalization, serialization, folder selection, and WorkDrive upload.
visaDisplay.js	Visa-label normalization and Zoho deal type mapping.
workDrivePreviewUrl.mjs	Convert supported public WorkDrive links to embed URLs.
zohoClient.js	Zoho token, CRM, related-list, WorkDrive, and attachment client.

10.3 Local UI component layer

src/components/ui contains local wrappers for alerts, badges, buttons, cards, checkboxes, collapsibles, dialogs, inputs, labels, progress, radio groups, scroll areas, selects, textareas, toasts, and the toaster. Admin managers compose these primitives rather than introducing a separate component framework.

11. Development, deployment, and operations

11.1 Local commands

```
npm install
npm run dev
npm run lint
npm run build
npm run start
node --test src/lib/pdfUploadRules.test.mjs src/lib/workDrivePreviewUrl.test.mjs
```

Package scripts

Command	Purpose
npm run dev	Start the Next.js development server.
npm run build	Create the production Next.js build.
npm run start	Run the production server after a build.
npm run lint	Run ESLint with Next core-web-vitals rules.
npm run migrate:shared-resources	Migrate legacy per-application resources to the global resources collection.

11.2 Shared-resource migration

scripts/migrate-shared-resources.js scans every application resources subcollection and copies eligible records to the global resources collection. It records migration.legacyResourcePath to avoid copying the same source twice.

- --dry-run reports work without writing.
- --include-archived includes archived source resources; they are skipped by default.
- Writes are committed in batches of 200.
- The script reads .env and .env.local and supports the same Firebase credential fallbacks.

11.3 Firestore deployment configuration

- Default database in region asia-east2.
- Rules file: firestore.rules.
- Indexes file: firestore.indexes.json.
- Tracked composite indexes cover messages by userId and createdAt ascending/descending.

Deployment note: The repository does not encode one production hosting provider. Next.js can be deployed to a compatible Node.js platform; Firebase configuration in this repository is specifically for Firestore rules and indexes.

11.4 Operational behavior

Operational characteristics

Scenario	Behavior
Zoho unavailable	CRM helper often returns null/empty data when no token is available; WorkDrive mutations throw because file operations cannot continue.
Duplicate matter IDs	Resolver selects one candidate deterministically and returns duplicateCount and duplicateIds.
Remote cleanup failure	Some upload flows delete the new WorkDrive resource when a later critical step fails.
Document replacement	Uploading a replacement adds a new active resource; it does not automatically archive all prior review resources.
Token lifetime	Process-memory cache is lost on restart/serverless cold start; the broker is called again.

12. Verification results, limitations, and risks

12.1 Repository checks performed for this document

Validation status on 23 July 2026

Check	Result
ESLint	Passed with 0 errors and 4 warnings.
Utility tests	2 passed: PDF upload rules and WorkDrive embed URL.
Warnings	Three next/no-img-element warnings and one anonymous default-export warning.
Source scope	All maintained files under src, scripts, root configuration, Firestore rules/indexes, and current package metadata were inspected.
Working tree	Pre-existing uncommitted application changes were present and were not modified by documentation generation.

12.2 Known implementation risks

Prioritized risk register

Risk	Evidence and impact
Critical - authorization bypass	getAdminSession always returns an admin actor, so login and admin guards do not enforce identity. Matter and comment endpoints also have no gate.
High - public file links	WorkDrive links are created as external/public links with downloads enabled. Access depends on possession of the URL.
High - legacy file orphaning	DELETE /api/resources/{id} deletes only Firestore metadata and does not remove the associated WorkDrive resource.
Medium - notification write defect	The review-comment POST path attempts to use appld when creating a non-document-review notification, but appld is not defined in that scope. The notification error is caught, so comment creation can still succeed without a notification.
Medium - replacement history	A document-review replacement creates another active resource. Selection favors the newest resource, but older active files remain until explicitly archived.
Medium - production logging	The Zoho client logs token response structures and sample CRM records, which can expose sensitive operational data in server logs.
Medium - rules mismatch for direct clients	Tracked Firestore rules do not define client access for global resources, resourceTemplates, notifications, or reviewComments. Server Admin SDK calls still work.
Low - stale auxiliary documentation	Some existing Markdown guides describe routes that are not present in the current source. Use this source-derived document and the route tree as the current baseline.

12.3 Intentional current limits

- Individual resource Embed, Library, and Folder actions are disabled.
- Visa-template category folders do not create WorkDrive subfolders.
- WorkDrive child scanning stops after 1,000 items per folder.
- Questionnaire answers are read-only; the portal is a review surface, not the intake editor.
- The generic print export uses the browser print dialog rather than a server PDF generator.

- The admin corrections dashboard aggregates Firestore comments only; Zoho Corrections are merged on the matter document-review screen.

Appendix A. Representative data shapes

A.1 Matter API response

```
{
  "success": true,
  "application": { "id": "firebase-app-id", "...": "matter metadata" },
  "questionnaire": { "...": "stored answers" },
  "completion": { "section-key": true },
  "percentage": 0,
  "progress": { "completedSections": 0, "totalSections": 0 },
  "matchedBy": "firebaseId | zohoId | zohoDealId | dealId",
  "duplicateCount": 1,
  "duplicateIds": ["firebase-app-id"]
}
```

A.2 Review comment

```
{
  "path": "profile:123.details.family_name",
  "label": "Family name",
  "body": "Please confirm the spelling.",
  "severity": "suggestion",
  "source": "questionnaire | documentReview",
  "status": "open | resolved",
  "sectionKey": "profile:123",
  "authorId": "admin | client",
  "authorName": "Admin | Client",
  "createdAt": "timestamp",
  "updatedAt": "timestamp"
}
```

A.3 Template item

```
{
  "parentId": null,
  "kind": "folder | file | link | note",
  "name": "Resource name",
  "category": "Guides",
  "order": 0,
  "status": "active | hidden",
  "externalUrl": null,
  "workdriveId": null,
  "mimeType": null,
  "size": null
}
```

A.4 Matter resource

```
{
  "type": "file | link | note",
  "title": "Resource title",
  "description": "",
  "status": "active | archived",
  "source": "documentReview (optional)",
  "category": "Document Review (optional)",
  "publicUrl": "https://...",
  "workDriveResourceId": "resource-id",
  "workDriveFolderId": "folder-id",
  "createdAt": "timestamp",
}
```

```
"updatedAt": "timestamp"
}
```

Appendix B. Source map

Primary evidence used

Documentation area	Repository source
Features and pages	src/app/**/page.js, src/app/matter/[matterId]/layout.js, src/components/admin/*.jsx
Internal APIs	src/app/api/**/route.js
Matter resolution	src/lib/matterResolver.js
Questionnaire structure/progress	src/lib/questionnaireSections.js, questionnaireProgress.js, routes.js
Firebase	src/lib/firebase-admin.js, firestore.rules, firestore.indexes.json, firebase.json
Zoho CRM/WorkDrive	src/lib/zohoClient.js, sharedResources.js, resourceTemplates.js
Authentication	src/lib/adminSession.js, src/app/api/admin/session/route.js
Dependencies	package.json, package-lock.json, pnpm-lock.yaml
Migration	scripts/migrate-shared-resources.js
Configuration	next.config.mjs, postcss.config.mjs, jsconfig.json, eslint.config.mjs

Maintenance rule: Update this documentation when route files, Firestore paths, Zoho field names, WorkDrive folder ownership, authentication behavior, or package versions change.